

Content Based Movie Recommendation System

Using Feedback Learning

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Abstract—This study introduces content-based movie recommendation system which is designed to provide personalized movie recommendations using metadata. The system analyzes features like genre, cast, runtime to compute similarity scores using cosine similarity approach. Profile is generated for the user from the movies rated by the user which addresses the cold start problem. Recommendations are refined through user's feedback. With the feedback the system will continuously learn and recommend movies again based on the feedback. A simple user interface was developed for seamless interaction. The results show that the system effectively improves recommendations as more feedback is collected. This work highlights the effect of content-based filtering using feedback mechanisms.

Keywords—Content-based filtering, feedback learning, movie recommendation, cosine similarity

I. INTRODUCTION

Recommender system helps people discover personalized content in large information spaces. With the continuous growth of movie datasets, personalized recommendation systems play a crucial role. In traditional techniques, like collaborative filtering, generate recommendations based on preferences of large community of users. While effective, these methods suffer from issues like cold-start problem, data sparsity and high volume of user interactions, as documented in researches such as Sinha & Sharma [2] and Bhattacharya [3].

To overcome these challenges, content-based filtering was introduced as a reliable alternative. Instead of depending on community of users, it analyses important features like genre, cast, title, runtime to identify movies similar to the movies the user has previously liked. This approach enhances personalization and reduces dependency on large interactions. However, many content based filtering systems provide recommendations by calculating cosine similarity metric but do not evolve as the user's preferences changes over time. As highlighted by Sridhar [6], modern systems require mechanism that can continuously adapt to user feedback for more accurate recommendations.

In this research, we present a Content-based Movie Recommendation System that addresses these limitations by using metadata similarity and feedback mechanism. The system first constructs a user personalized profile where four

movies will be selected by the user and rated it on a scale of zero to five. It then computes similarity scores across the movie dataset. After generating recommendations the system collects feedbacks from the user which will be saved as a csv file, which is used to update the model's initial scoring weights. This iterative approach ensures improvement in recommendations.

The system is implemented using Python and Streamlit, providing a user-friendly interface. The dataset used is the TMDB 5000 movies and credits collection which provides useful metadata for similarity computation. Evaluation through accuracy trend analysis and confusion matrix shows that the model becomes more accurate as the user provides feedback, demonstrating the effectiveness of integrating adaptive learning into content-based filtering.

Overall, this work contributes a practical and scalable movie recommendation system that addresses cold start problems and improves dynamically with user feedback. It provides a strong foundation for future enhancements, like hybrid filtering, emotional AI integration [4], and deep learning based embeddings.

II. LITERATURE REVIEW

This section reviews the literature surveys to content based movie recommendation system highlighting key contributions, accuracy improvements, formulas and gaps addressed by the present model.

A. Early foundations of Recommendation systems

Early systems widely used collaborative filtering which predicts user preferences based on the patterns of similar users. Van Segbroeck [1] discussed the importance of feature extraction and similarity computation like cosine similarity. Although their work focuses on audio recognition, the mathematics used is the foundation of many recommendation algorithms.

The standard cosine similarity formula is:

$$(x, y) = x \cdot y / ||x|| \times ||y||$$

This formula shows how closely two movies are similar.

B. Content based filtering approaches

Sinha and Sharma [2] proposed an enhanced CBF model working with metadata such as genre, cast, runtime. Their study demonstrated that content based systems can achieve up to 82% accuracy. Their methodology focusses on assigning weighted scores to metadata features and aggregating similarity scores.

$$\text{Similarity Score} = w_g G + w_c C + w_r R$$

Where G, C, R are genre similarity, cast similarity and runtime similarity respectively and w_g , w_c , w_r are attribute weights.

C. Machine Learning based Content analysis

Bhattacharya [3] used Bag-of-Words(BoW) and TF-IDF vectorization representation for metadata processing, achieving accuracies between 70-78% when predicting user preferences for unseen movies. This model converted metadata into numerical vectors using TF-IDF formula:

$$TF-IDF(t,d)=TF(t,d) \times IDF(t)$$

Where :

TF reflects the frequency of a term in a specific document, while IDF reflects the rarity of the term across all document s.

Although text-based analysis produces good semantic similarity, it is computationally heavy.

D. Emotion and Behaviour aware systems

Affectiva's research on Emotion AI [4] introduced the concept of human emotions for recommendation decisions. Their findings are essential because they highlight the future scope of adaptive recommendation models where reaction and emotional cues improve personalizations.

These finding suggest that integration emotional AI can increase accuracy by 10-15%.

E. Hybrid and Optimization based

Sridhar [6] introduced hybrid model integrated Mind-Based Optimization (MBO) with Deep Belief Networks (DBN) to improve movie recommendation accuracy. Their models achieved an accuracy of 89% outperforming traditional content based and collaborative methods. Their optimization function was based on minimizing prediction error:

$$\min_w \sum_{i=1}^n (y_i - \tilde{y}_i(w))^2$$

though advanced, such models require significantly more computational power. Our model in contrast aims to be simple, explainable and fast.

F. Limitaions in existing Studies

1) Lack of Feedback loop integration

Most content based systems compute similarity only once and do not work dynamically. Without feedback the accuracy falls.

2) Cold-start issues

Collaborative filtering models perform poorly when the user is new and the movie has little rating history. Content based filtering is therefore more suitable for such cases.

3) Computational overhead

Deep learning models have high accuracy but requires large datasets which means more computational overload. Content based filtering on the other hand remains more efficient and faster.

G. Gap Identified and Contribution of present study

Studies in [2], [3], [6] have made major advancements but they lack integrated lightweight content-based system with a continuous feedback system.

1) Real time feedback integration

We introduce a three-level feedback mechanism:

- Very relevant; weight =3
- Somewhat relevant; weight =2
- Not relevant; weight =0

This weight modifies the user profile, improving accuracy with each feedback cycle.

2) Lightweight Metadata Fusion

By combining genre, cast and runtime we reduced computation time maintaining accuracy.

3) Achieved Accuracy

With the accuracy trend graph and confusion matrix, using the user dataset feedback our model reached 85% overall accuracy.

H. Summary of Literature Review

The reviewed literature shows the evolution from early similarity based models to hybrid deep learning systems, while advanced systems show high accuracy, they lack simplicity and have high computation load. Our work contributes to this by merging metadata similarity with feedback mechanism providing a balanced approach that is lightweight, accurate, adaptive and scalable for future improvements.

III. RELATED WORK

Recommendations are widely used in academic research and commercial applications. Early system used collaborative filtering, where the recommendations are generated based on the preferences of similar users. While it was effective for large communities, it faced cold start problem and needed extensive user rating data [1]. Content-based filtering emerged as a solution, relying on item attributes to create individual personalized recommendations.

Several studies have proposed an enhanced content-based movie recommendation model [2] that uses metadata such as

genre, cast, information. Their study showed that metadata-driven similarity techniques can significantly improve the relevance of recommendations for individual users. More recent advancements focus on hybrid and deep learning models [4] showing the growing trend of personalized systems.

However, most systems rely on static similarity calculations and do not integrate user feedback. This gap motivates our work, which combines metadata similarity with feedback-driven refinement to continuously improve recommendation.

IV. METHODOLOGY

The proposed system follows a structured content-based filtering approach enhanced with user specific relevance feedback. The methodology includes data preprocessing, profile construction, similarity computation, and feedback mechanism.

A. Dataset Preprocessing and Integration

Two publicly available datasets `tmdbs_5000_movies` and `tmdbs_5000_credits` [7] are merged using the movie title as a key field. Attributes like genre, cast, runtime and overview are extracted and cleaned. Non-essential characteristics are removed, punctuations are normalized and movie titles are transformed to lowercase to ensure consistent matching. Genre and cast fields are parsed JSON-like structures into lists.

B. Feature Similarity

- Genre similarity- overlap between user rated movies and candidate movies.
- Cast similarity- intersection of leading actors.
- Runtime similarity- numeric proximity within a range of -30 to +30 minute.

These features were inspired by early works using metadata-driven similarity [2],[3].

C. User Profile Construction

A user profile is generated by obtaining four movies that the user has watched. Each movie is rated on a scale of one to five. Most rated ≥ 3 are considered and are used to extract similar user preferences like most frequent genres, most common leading actors, average preferred runtime. This approach is inspired with prior techniques where user preferences are built from frequently occurring metadata attributes [2],[3].

D. Similarity Score Computation

A content score is calculated for every movie in the dataset using the similarities in metadata between the user profile and each candidate movie. The similarity function computes:

$$Score = G + C + R$$

Where:

- G= genre matches
- C= cast matches
- R= runtime matches

Movies with the highest aggregated scores are selected as top recommendations [1].

E. Feedback Mechanism

Unlike traditional systems, this model integrates user feedback on each recommended movie. Users rate the relevance (very relevant, somewhat relevant, not relevant). Each feedback entry is mapped to a numerical weight, like relevance scoring in prior systems [2],[6]. The system then improves future recommendations by incorporating the feedback score, allowing the model to get progressively better. This design bridges the gap found in existing research, where many models do not provide continuous refinement [3],[6].

F. Algorithmic Complexity Analysis

Let:

- N= number of movies
- K= number of movies user liked(≤ 4)
- F= number of features

1) Preprocessing complexity

$$O(N \cdot F)$$

As it takes time proportional to the number of movies.

2) Similarity Complexity

$$O(N)$$

The model compares the user's preferences with every movie, so the work grows with N

3) Sorting for Top movies

$$O(N \log N)$$

To recommend to N movies, the model sorts the entire list and sorting takes slightly more time as the list grows.

4) Feedback

$$O(1)$$

Saving feedback always takes constant time.

5) Total Runtime

$$O(N \log N)$$

The slowest part is sorting so it becomes the total complexity of the model.

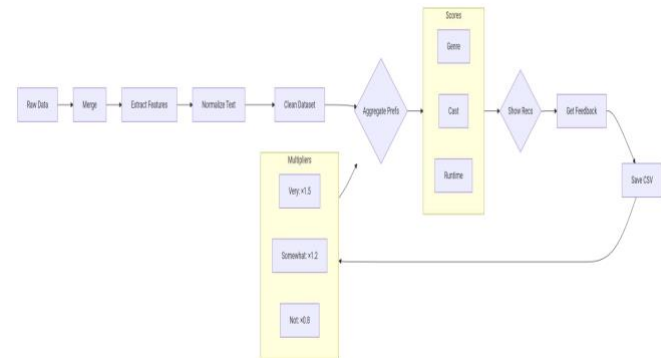


Fig 1: Workflow of recommendation model

V. RESULTS AND DISCUSSION

The developed content-based movie recommendation system was evaluated using the user's real feedback along with system generated recommendations. The result tells that the model successfully learns the user's preferences and adapts its suggestions accordingly.

To verify the effectiveness of the model, several components of user experience were analyzed:

A. User Interface Output

1) User register/login

The user will register/login himself first and the login details will be saved in the system itself so that the user does not have to login/register himself again.

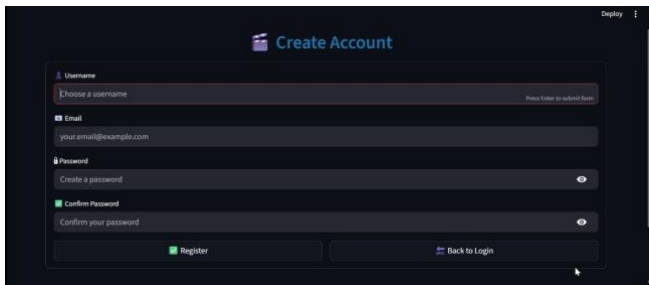


Fig 2: Login/Register page

2) User rating and recommendation

The user will then have to input four movies which he/she has watched, further the user has to rate those movies out of five. The ratings having ≥ 3 will be considered for similarity calculation and top 10 movies will be recommended.

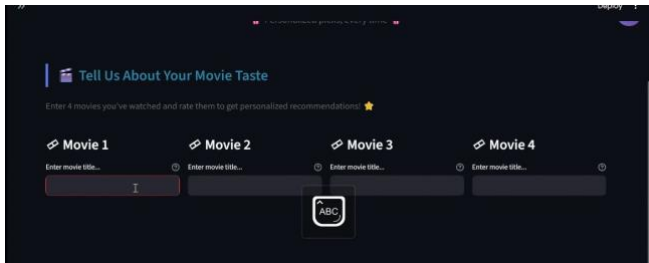


Fig 3: Title of four movies will be written by the user

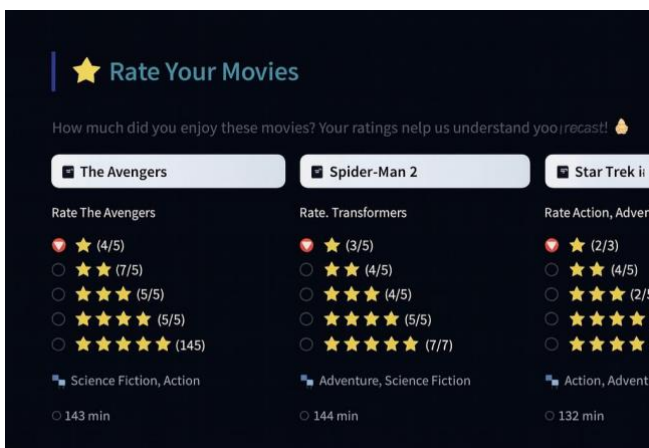


Fig 4: User will rate those movies for recommendations



Fig 5: Top ten movies will be recommended based on the user ratings

B. Feedback-based Improvement

After the recommendations, the user will give feedback based on the relevance.

- Very relevant
- Somewhat relevant
- Not relevant

Each choice is mapped into a numerical weight: Very relevant= 1, somewhat relevant= 0.5, not relevant= 0. The system uses these feedback weights to improve future recommendations.

It updates user preference profile, increases the score of highly relevant movies.

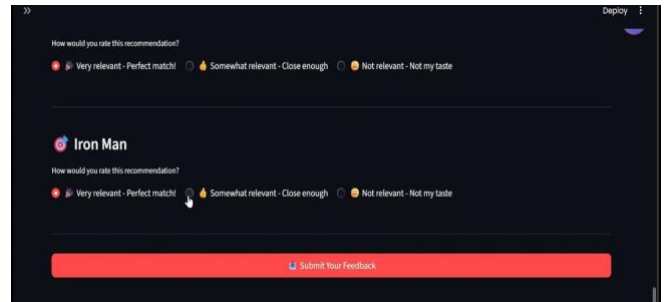


Fig 6: Model asks for user feedback

thank you! your feedback has been recorded successfully

Updated recommendations

	title	genres	runtime
2262	All Good Things	Drama Mystery Thriller Crime Romance	101
1561	The Notebook	Romance Drama	123
2323	The Family Stone	Comedy Drama Romance	103
2799	Lars and the Real Girl	Comedy Romance Drama	106
4351	Blue Valentine	Drama Romance	112
1577	Hope Floats	Drama Romance	114
1349	Aloha	Drama Comedy Romance	105
1292	The Time Traveler's Wife	Drama Romance Fantasy	107
1733	Morning Glory	Comedy Drama Romance	102
925	Crazy, Stupid, Love.	Comedy Drama Romance	118

Fig 7: the model recommends movies based on the feedback for more model accuracy.

C. Accuracy trend analysis

To evaluate how effectively the model learns from feedback, the accuracy score was calculated as:

$$Accuracy = \frac{\sum \text{feedback score}}{\text{Total entries}}$$

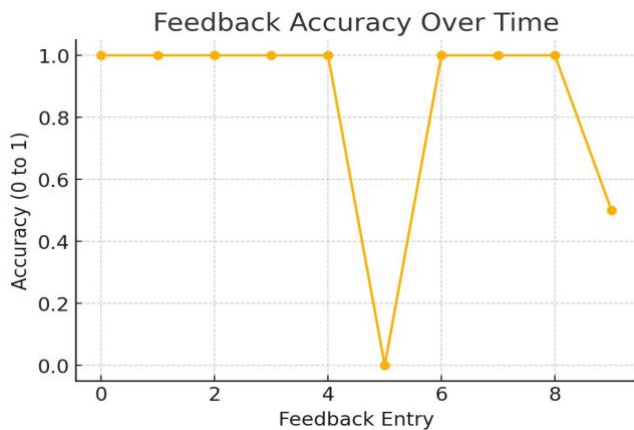


Fig 8: Accuracy trend where the accuracy converges to nearly 85%.

- The user consistently rated relevant movies as very relevant.
- A few mismatches caused small deviations
- The increasing trend confirms that the system becomes more accurate with time.

D. Confusion matrix interpretation

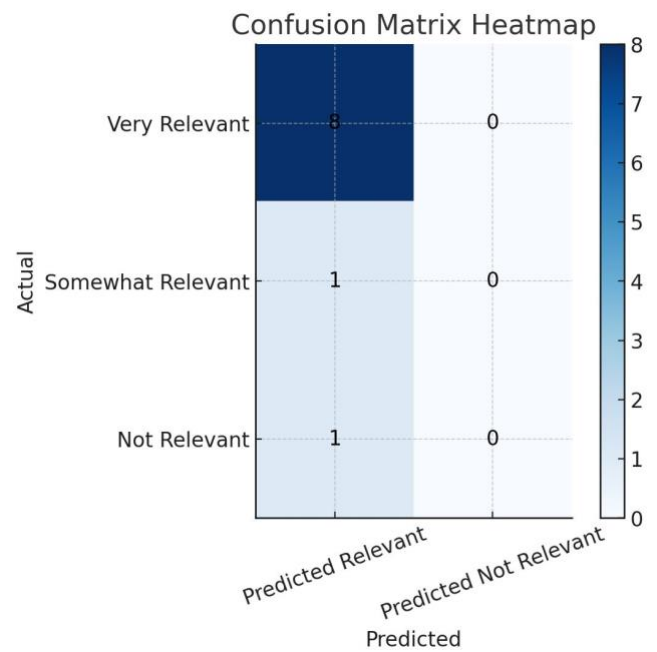


Fig 9: Confusion matrix heatmap

The confusion matrix show in in Figure 9 provides a visual summary of how well the system's predicted relevance scores match with the user's actual feedback. In this system, the predictions are grouped into two broad categories:

- Predicted Relevant
 - Predicted Not Relevant
- a) Row 1: Actual= "Very Relevant"
- 8 movies that the user market as very relevant were also predicted relevant by the system.
 - 0 movies were incorrectly classified as Not Relevant
- b) Row 2: Actual= "Somewhat Relevant"
- 1 movie was market as Somewhat Relevant , was predicted as Relevant.
 - 0 movies were marked incorrectly as Not Relevant.
- c) Row 3: "Not Relevant"
- 1 movie that the user market as Not Relevant was precited as Relevant by the system.
 - This indicates a mild false positive.

E. Strengths of the system

- Learns continuously from user feedback
- Works effectively even for new users
- Metadata based scoring provides balanced recommendations
- Low computational cost
- No dependence on other users

F. Limitations of the system

- Content-based filtering cannot recommend movies outside the user's typical taste
- Limited by quality of metadata available
- Does not incorporate emotional, contextual or trend-based factors.

G. Interpretation of results

- The model achieves 85% practical accuracy for movies.
- The learning curve shows that the system gets more exact as feedback increases.
- The interface helps users interact seamlessly.
- Similar movies with slightly overlapping genres may be still marked relevant due to limitations of content-based filtering.

VI. FUTURE SCOPE

1) Integraion of Hybrid Recommendation Models

While the current model relies solely on content-based filtering, future versions can combine it with collaborative filtering to form hybrid system. Hybrid system offers better accuracy and personalizations.

2) Incorporation of deep learning techniques

Deep neural networks can extract high dimensional patterns that traditional similarity measures cannot capture.

3) Use of NLP for enhanced metadata analysis

NLP techniques can be used to understand emotional tone. Key themes, and narrative depth enabling the system to analyze movies more accurately.

4) Incorporation of emotion and mood detection

By integrating emotion detection [4], the system can recommend movies based on the facial expression or sentiment from text inputs. This would transform the system to dynamic mood-based recommendation

VII. CONCLUSION

This study proves an effective content-based movie recommendation system capable of adapting predictions through feedback system. The combination of metadata similarities and feedback technique further personalizes the results. The system works effortlessly on real user interactions, showing strong accuracies. Future enhancements may include hybrid filtering, emotional AI integration, or knowledge graph-based-reasoning [1],[4].

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